

Comparison of Projected Heun/RK2 and Incremental Adams–Bashforth 2

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1 Purpose

This private report compares the two implementations retained in the repository. Both solve the same non-dimensional, axisymmetric incompressible pipe-flow and heat-transfer problem on the same staggered finite-volume grid. They use identical spatial operators, boundary conditions, physical parameters, and convergence definitions. Only the temporal integration and pressure-projection organization differ.

The production assignment report uses the Adams–Bashforth 2 implementation. The Heun implementation is retained as an independent reference solution.

2 Common discretization

Both solvers use $n_r \times n_z = 50 \times 250$, $\Delta r = 0.01$, $\Delta z = 0.2$, $\Delta t = 0.002$, $\text{Re} = 100$, and $\text{Pr} = 5$. Convection is evaluated using second-order linear upwind differencing and diffusion using second-order central differences. Pressure and temperature are cell-centred; radial and axial velocity are stored on the corresponding control-volume faces.

For both methods, a projection solves

$$\nabla^2 \phi = \frac{1}{\Delta t} \nabla \cdot \mathbf{u}^*, \quad \mathbf{u}^{\text{new}} = \mathbf{u}^* - \Delta t \nabla \phi. \quad (1)$$

3 Runge–Kutta, RK2, and Heun terminology

Runge–Kutta methods are a family of time-integration methods for ordinary differential equations

$$\frac{d\phi}{dt} = F(t, \phi). \quad (2)$$

They advance the solution by evaluating the right-hand side F at one or more intermediate points, called *stages*, inside the current time step. These stage values are combined to approximate the time integral of F . Unlike Adams–Bashforth methods, a Runge–Kutta method normally does not need RHS values stored from earlier complete time steps.

RK2 means a Runge–Kutta method with second-order temporal accuracy. It is a family rather than one unique formula. Explicit midpoint, Ralston’s method, and Heun’s method are all RK2 schemes. They differ in where the second stage is evaluated and in the weights used to combine the stages.

Heun’s method is the explicit trapezoidal member of the RK2 family. It first uses an Euler prediction over the complete time step,

$$k_1 = F(t^n, \phi^n), \quad (3)$$

$$\tilde{\phi} = \phi^n + \Delta t k_1, \quad (4)$$

then evaluates the slope at the predicted endpoint and averages both slopes:

$$k_2 = F(t^{n+1}, \tilde{\phi}), \quad (5)$$

$$\phi^{n+1} = \phi^n + \frac{\Delta t}{2}(k_1 + k_2). \quad (6)$$

The average is analogous to the trapezoidal rule, which explains the name *explicit trapezoidal method*. Thus “Heun”, “explicit trapezoidal RK2”, and the particular two-stage equations above refer to the same method; “RK2” alone is less specific.

For incompressible flow, each velocity used to evaluate a stage should also satisfy the continuity constraint. The implementation therefore adds a pressure projection after Heun’s Euler predictor and another after the final weighted predictor. These projections are not part of generic Heun theory; they are how the time integrator is coupled to incompressibility.

4 Projected Heun/RK2 method

Let $F(\mathbf{u})$ contain the discrete convective and viscous momentum terms, but not pressure. The Heun implementation performs

$$k_1 = F(\mathbf{u}^n), \quad (7)$$

$$\hat{\mathbf{u}}_1 = \mathbf{u}^n + \Delta t k_1, \quad (8)$$

$$\mathbf{u}_1 = \mathcal{P}(\hat{\mathbf{u}}_1), \quad (9)$$

$$k_2 = F(\mathbf{u}_1), \quad (10)$$

$$\hat{\mathbf{u}}^{n+1} = \mathbf{u}^n + \frac{\Delta t}{2}(k_1 + k_2), \quad (11)$$

$$\mathbf{u}^{n+1} = \mathcal{P}(\hat{\mathbf{u}}^{n+1}), \quad (12)$$

where $\mathcal{P}$ denotes the pressure projection. Projecting the intermediate stage ensures that k_2 uses a divergence-free transport velocity. This requires two momentum RHS evaluations and two Poisson solves per complete step.

The old pressure is not included in the predictor. Consequently, the final Poisson unknown is the complete non-incremental projection pressure rather than a correction. The stored pressure is therefore blended toward it:

$$p^{n+1} = (1 - \alpha_p)p^n + \alpha_p \phi^{n+1}. \quad (13)$$

Temperature is advanced with the same explicit trapezoidal weights and uses the projected intermediate velocity in its second stage.

5 Incremental Adams–Bashforth 2 method

The AB2 method extrapolates the explicit RHS from two time levels:

$$F^{AB2} = \frac{3}{2}F^n - \frac{1}{2}F^{n-1}. \quad (14)$$

Because no previous RHS exists initially, the first step uses forward Euler. Subsequent predictors include the known pressure gradient:

$$\mathbf{u}^* = \mathbf{u}^n + \Delta t (F^{AB2} - \nabla p^n). \quad (15)$$

One Poisson solve produces the pressure correction p' , after which

$$\mathbf{u}^{n+1} = \mathbf{u}^* - \Delta t \nabla p', \quad p^{n+1} = p^n + \alpha_p p'. \quad (16)$$

Temperature uses AB2 on its convection–diffusion RHS. The method requires one current RHS evaluation, stored previous RHS arrays, and one Poisson solve per step.

6 Direct comparison

Table 1: Algorithmic differences per complete time step.

Property	Projected Heun/RK2	Incremental AB2
Temporal form	two-stage, one-step	two-level, multistep
Formal temporal order	second	second after startup
Startup	self-starting	first-order Euler first step
Momentum RHS evaluations	two	one
Poisson solves	two	one
Stored previous RHS	not required	required
Old pressure in predictor	no	yes
Poisson unknown	full projection pressure	pressure correction
Pressure update	relaxed replacement	relaxed addition

The AB2 formulation is less expensive per step because sparse pressure solutions dominate the computational cost. Heun is self-starting and its stage structure is often easier to generalize, but the correctly projected version doubles the Poisson work. AB2 also corresponds directly to the procedure recorded in the lecture notes.

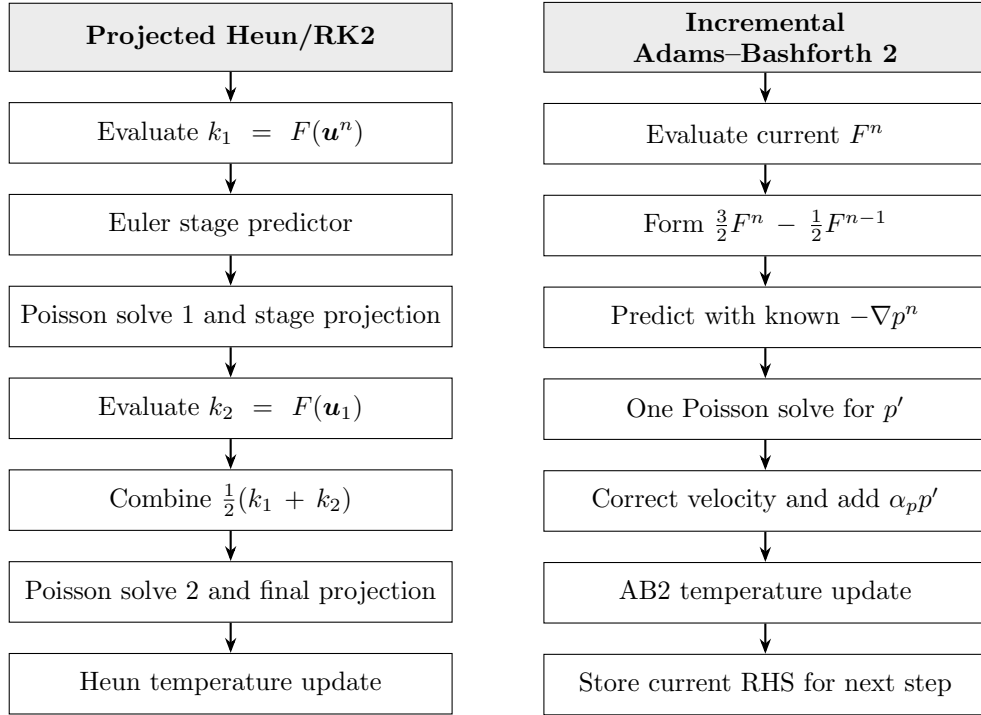


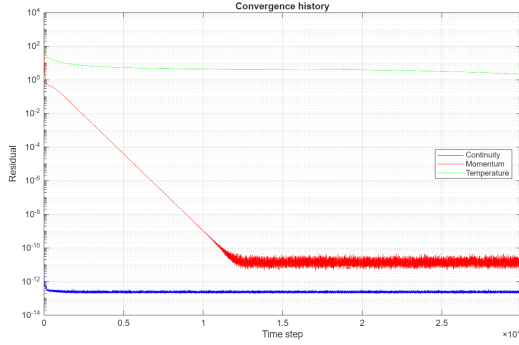
Figure 1: Per-step workflows. The first AB2 step substitutes forward Euler for the unavailable two-level extrapolation.

7 Measured results

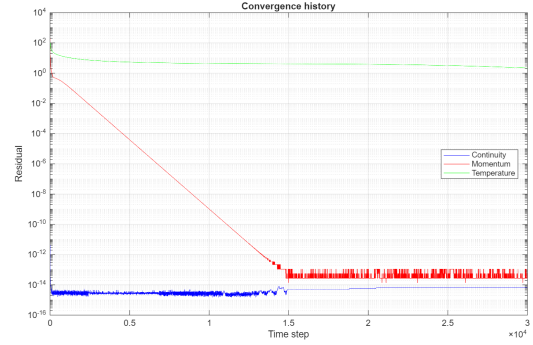
Both implementations were run to 30,000 steps, corresponding to $t^* = 60$. The AB2 results were independently generated in Python and MATLAB. The final reported residuals were

Run	R_{cont}	R_{vel}	R_T
AB2, MATLAB	6.43×10^{-15}	2.78×10^{-14}	2.19
AB2, Python	1.30×10^{-9}	5.55×10^{-14}	2.19
Heun reference	$\mathcal{O}(10^{-13})$	$\mathcal{O}(10^{-13})$	2.19

The different continuity floors in the two AB2 programs arise from numerical linear-solve precision and remain far below the required 10^{-6} tolerance. Velocity is converged in both methods. Temperature is not temporally converged at $t^* = 60$ in either method. Its identical residual of 2.19 demonstrates that changing from Heun to AB2 does not remove the slow physical thermal relaxation timescale.

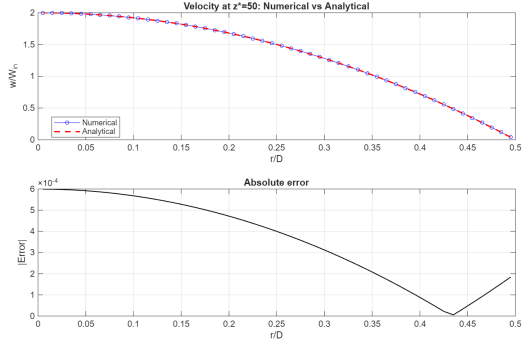


(a) Projected Heun/RK2.

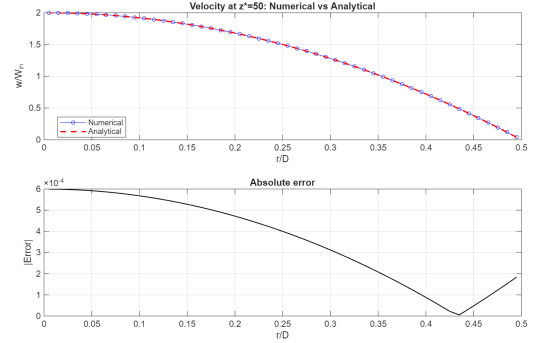


(b) Incremental AB2.

Figure 2: Convergence histories under identical physical and grid settings.



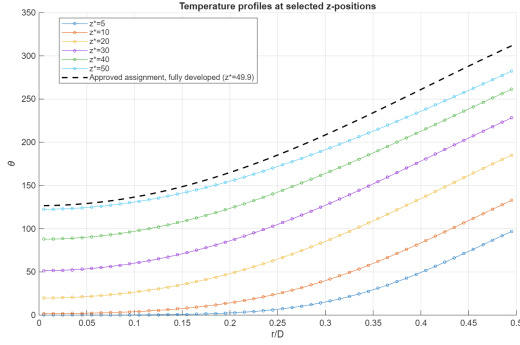
(a) Projected Heun/RK2.



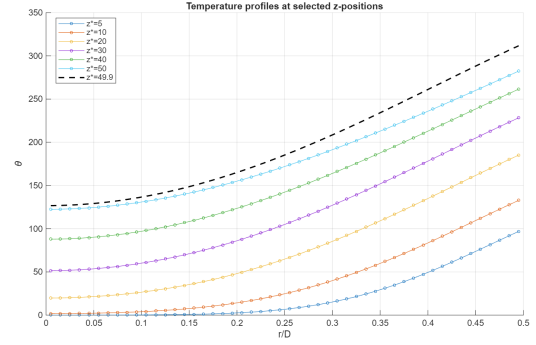
(b) Incremental AB2.

Figure 3: Fully developed axial velocity compared with Hagen–Poiseuille flow.

The two panels above are visually identical because both methods have reached the same discrete steady momentum solution. Time integration determines the transient path to that fixed point, but it does not change the converged finite-volume equations. With velocity residuals near machine precision, the remaining differences are below the plotted line width. Profiles saved at an early common time would be a more sensitive comparison of temporal behavior.



(a) Projected Heun/RK2.



(b) Incremental AB2.

Figure 4: Temperature profiles after 30,000 time steps.

8 Conclusion

For this steady laminar test case, both second-order methods produce the same engineering conclusions and effectively identical convergence behavior. AB2 is preferable for the assignment because it follows the documented lecture algorithm and halves the number of Poisson solves per step. The Heun solver remains useful as an independent cross-check. Neither method accelerates the physical approach of the temperature field to steady state; longer simulated time or a steady/implicit thermal treatment would be required for that.